PHYSICS

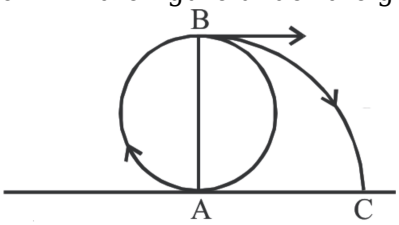
1) Two particles A and B are projected from ground with speed 2 m/s and $\frac{4}{\sqrt{3}}$ m/s from horizontal at an angle 30° and 60° respectively. If separation between A and B on ground is 10 m. Find relative angular velocity of B w.r.t A :

(1) 0.1 rad/s
(2) 0.2 rad/s
(3) 0.05 rad/s
(4) 1 rad/s

2) The kinetic energy K of a particle moving along a circular path of radius r depends upon the distance s as $K = As^2$. The force acting on the particle is :-

(1) 2 As
(2) $2A\sqrt{r^2+s^2}$
(3) $2As\sqrt{1+\frac{s^2}{r^2}}$
(4) 2 Ar

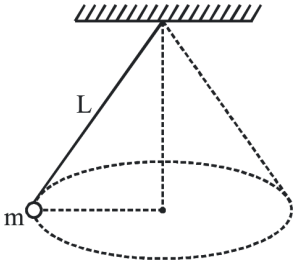
3) An object is tied to a string of length l and is revolved in a vertical circle at the minimum velocity. When the object reaches the upper most point, the string breaks and it describes a parabolic path as shown in the figure under the gravitational force. The horizontal range AC in the plane of A would be :-



(1) x = 1
(2) x = 2
(3) $x = \sqrt{2}\ell$
(4) $x = 2\sqrt{2}\ell$

4) A ball of mass (m) 0.5 kg is attached to the end of a string having length (L) 0.5 m. The ball is rotated on a horizontal circular path about vertical axis. The maximum tension that the string can bear is 324 N. The maximum possible value of angular velocity of ball (in radian/s) is :-

AIM : FREE EDUCATION TO ALL ♥ APUL



(1) 9

(2) 18

(3) 27

(4) 36

5) Keeping the angle of banking same, if the radius of curvature is made four times, the percentage increase in the maximum speed with which a vehicle can travel on a circular road is :-

(1) 25%

(2) 50%

(3) 75%

(4) 100%

6) A circular road of radius 1000 m has banking angle 45°. What will be the maximum safe speed of a car having mass 2000 kg, if the coefficient of friction between tyre and road is 0.5 ?

(1) 172 m/s

(2) 124 m/s

(3) 99 m/s

(4) 86 m/s

7) When a ceiling fan is switched off its angular velocity reduces to 50% while it makes 36 rotations. How many more rotation will it make before coming to rest (Assume uniform angular retardation):

(1) 18

(2) 12

(3) 36

(4) 48

8) A stone of mass m is tied to a string and is moved in a vertical circle of radius r making n revolutions per minute. The total tension in the string when the stone is at its lowest point is

(1) mg

(2) $m(g + \pi n r^2)$

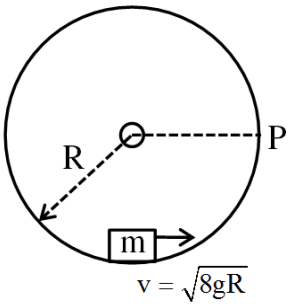
(3) $m(g + \pi n r)$

(4) $m\{g + (\pi^2 n^2 r) / 900\}$

9) A block of mass m is performing circular motion inside the surface of a circlar path. Its velocity at

AIM : FREE EDUCATION TO ALL ♥ APUL

AIM : FREE EDUCATION TO ALL ♥ APUL



the bottom is $\sqrt{8gR}$. Which of the followings are correct

(i) Maximum normal force is 9mg

(ii) Minimum velocity is at the top and it is $\sqrt{2gR}$

(iii) Minimum normal force is 0

(iv) Centripetal acceleration when block is at horizontal position P is 6 g

(1) only (i), (iv) are correct

(2) only (i), (ii), (iii) are correct

(3) only (i), (ii), (iv) are correct

(4) only (ii), (iii), (iv) are correct

10) A particle is moving along a circular path. The angular velocity, linear velocity, angular acceleration and centripetal acceleration of the particle at any instant respectively are $\vec{\omega}, \vec{v}, \vec{\alpha}$ and $\vec{a}_c$. Which of the following relations is not correct ?

(1) $\vec{\omega} \perp \vec{v}$

(2) $\vec{\omega} \perp \vec{\alpha}$

(3) $\vec{\omega} \perp \vec{a}_c$

(4) $\vec{v} \perp \vec{a}_c$

11) A particle of mass 'm' describes a circle of radius r. The centripetal acceleration of the particle is $\frac{2}{r^3}$. The momentum of the particle is :-

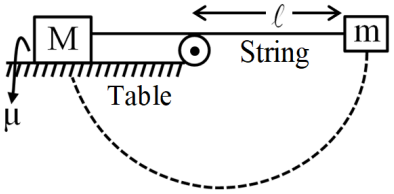
(1) $\frac{\sqrt{2}m}{r}$

(2) $\frac{2m}{\sqrt{r}}$

(3) $\frac{4m}{r}$

(4) $\frac{4m}{\sqrt{r}}$

12) A block of mass m is released from the horizontal position, as shown in the diagram. What should be the minimum value of coefficient of friction between block of mass M and surface of the table, to



avoid slipping of block of mass M on table ?

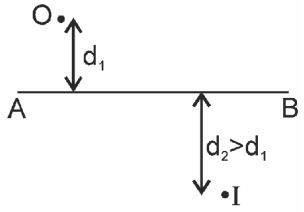
(1) $\frac{3M}{m}$

(2) $\frac{M}{2m}$

(3) $\frac{2m}{M}$

(4) $\frac{3m}{M}$

13) In the figure shown, the image of a real object is formed at point I. AB is the principal axis of the



mirror. The mirror must be :

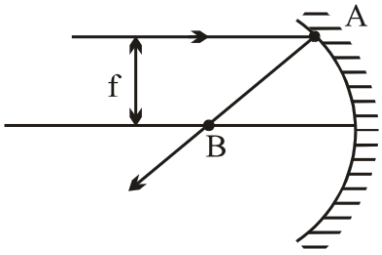
(1) concave and placed towards right of I

(2) concave and placed towards left of I

(3) convex and placed towards right of I

(4) convex and placed towards left of I.

14) A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point B from the centre of curvature to the focal length is (AB is the reflected ray) :-



(1) $\frac{2}{\sqrt{3}}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{2}{3}$

(4) $\frac{1}{2}$

15) An object is placed at 30 cm from concave mirror. If the image is real & 2 times magnified, then find the another location of object such that image is again 2 times magnified :-

(1) –20 cm

(2) –30 cm

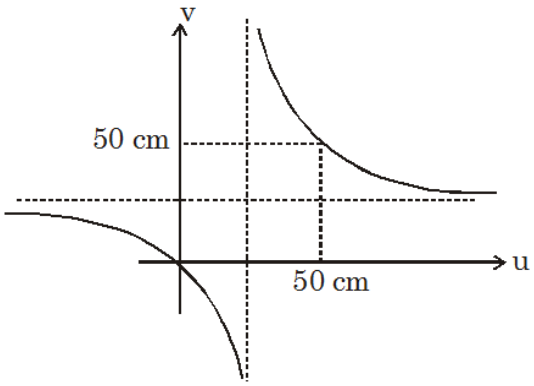
(3) –10 cm

(4) –15 cm

AIM : FREE EDUCATION TO ALL ♥ APUL

AIM : FREE EDUCATION TO ALL ♥ APUL

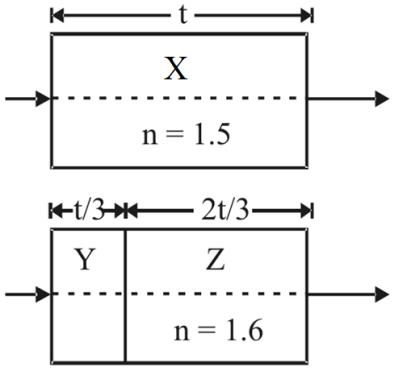
16) u-v graph for a convex mirror is shown in figure. A real object placed 25cm away from mirror



will have transverse magnification :

(1) 2
(2) 0.75
(3) 0.5
(4) –0.5

17) Two transparent slabs have the same thickness. One is made of material X of refractive index 1.5. The other is made of two material Y and Z having thickness in the ratio 1 : 2. The refractive index of Z is 1.6. If a monochromatic parallel beam passing through the slabs has the same number

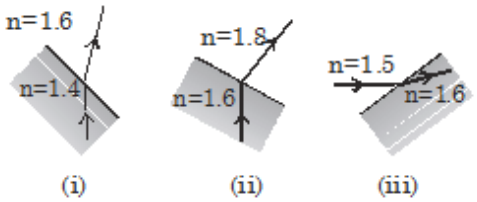


of wavelengths inside both, the refractive index of Y is -

(1) 1.1
(2) 1.2
(3) 1.3
(4) 1.4

18)

Which of the following ray diagram shows physically possible refraction ?



(1) (i)
(2) (ii)
(3) (iii)

(4) None of these

19) A beam of light composed of red and green rays is incident obliquely at a point on the face of a rectangular glass slab. When coming out on the opposite parallel face, the red and green rays emerge from-

(1) Two points and in same direction

(2) One point and in two different directions

(3) One point and in the same direction

(4) Two points and in two different non parallel directions

20) When a small lamp is held 1.5m above the surface of water in a tank, its image formed by reflection at the surface appears to coincide with the image of the bottom of the tank. (μ of water = 4/3). The depth of the tank is

(1) 2 m

(2) 1.5 m

(3) 1 m

(4) 4 m

21) If light travels a distance x in t_1 sec in air and 10x distance in t_2 sec in a medium, the critical angle of the medium will be:

(1) $\tan^{-1}\left(\frac{t_1}{t_2}\right)$

(2) $\sin^{-1}\left(\frac{t_1}{t_2}\right)$

(3) $\sin^{-1}\left(\frac{10\,t_1}{t_2}\right)$

(4) $\tan^{-1}\left(\frac{10\,t_1}{t_2}\right)$

22) A ray of light is incident at an angle of 60° on one face of a prism which has an angle of 30°. The ray emerging out of the prims makes an angle of 30° with the incident ray. Calculate refractive index of the material of prism.

(1) 1

(2) $\sqrt{3}$

(3) $\sqrt{2}$

(4) $\frac{\sqrt{3}+1}{\sqrt{2}}$

23) The refracting angle of prism is A and refractive index of material of prism is $\cot\frac{A}{2}$. The angle of minimum deviation is :-

(1) 180° –3A

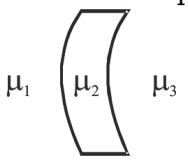
AIM : FREE EDUCATION TO ALL ♥ APUL

(2) 180° + 2A

(3) 90° –A

(4) 180° – 2A

24) Focal length of the lens is nR as shown in the figure. If the radii of curvature of the both the surfaces are equal to R, then find value of n, if an object is placed at infinity ($\mu_1 = 1$, $\mu_2 = 0.5$, $\mu_3 = 2$)



:-

(1) 1

(2) 2

(3) 1.5

(4) 2.5

25) A point object is at 30 cm from a convex lens ($\mu = 3/2$) of focal length 20 cm. The final image of object will be at infinity, if :-

(1) another concave lens of focal length 60 cm is placed in contact with the previous lens

(2) another convex lens of focal length 60 cm is placed at a distance of 30 cm from the first lens

(3) the whole system is immersed in a liquid of refractive index 4/3

(4) the whole system is immersed in a liquid of refractive index 2

26) A lens behaves as a converging lens in air and a diverging lens in water ($\mu = 1.33$). The refractive index of the lens material is-

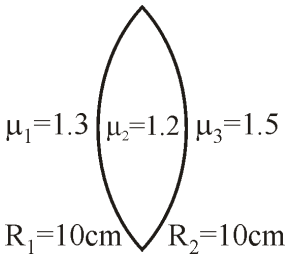
(1) Equal to unity

(2) Equal to 1.33

(3) Between unity and 1.33

(4) Greater than 1.33

27) Power of thin lens shown below is :-



(1) 4D

(2) –4D

(3) +2D

(4) –2D

28) In the displacement method , a convex lens is placed in between an object and a screen. If one of the magnification is 3 and the displacement of the lens between the two positions is 24 cm, then the focal length of the lens is :-

AIM : FREE EDUCATION TO ALL ♥ APUL

(1) 10 cm

(2) 9 cm

(3) 6 cm

(4) 16/3 cm

29) In a compound microscope, the focal length of the objective and the eye lens are 2 cm and 5 cm respectively. An object is placed at 3 cm before the objective and image is formed at the least distance of distinct vision, then the distance between two lenses (i.e. length of the microscopic tube) will be :-

(1) 10.16 cm

(2) 18.67 cm

(3) 23.00 cm

(4) 52.00 cm

30) Magnifying power of an astronomical telesope is 5. The separation between objective and eye-piece is 36 cm and the final image is formed at infinity. Calculate focal length of objective lens :-

(1) 30 cm

(2) 25 cm

(3) 6 cm

(4) 12 cm

31) A concave mirror of radius of curvature 30 cm, forms image of size three times that of object. The position(s) of the object may be at a distance from mirror equal to :-
(i) 10 cm (ii) 20 cm (iii) 40 cm

(1) only (ii)

(2) only (i)

(3) (i), (iii)

(4) (i), (ii)

32) A green light is incident from the water to the air - water interface at the critical angle (θ). Select the correct statement.

(1) The spectrum of visible light whose frequency is more than that of green light will come out to the air medium.

(2) The entire spectrum of visible light will come out of the water at various angles to the normal

(3) The entire spectrum of visible light will come out of the water at an angle of 90° to the normal.

(4) The spectrum of visible light whose frequency is less than that of green light will come out to the air medium.

33) The focal length of a thin biconvex lens is 20 cm. When an object is moved from a distance of 25 cm in front of it to 50 cm, the magnification of its image changes from m_{25} to m_{50}. The ratio $\frac{m_{25}}{m_{50}}$ is

(1) 2

AIM : FREE EDUCATION TO ALL ♥ APUL

(2) 4

(3) 6

(4) 8

34) A parallel beam of fast moving electrons is incident normally on a narrow slit. A fluorescent screen is placed at a large distance from the slit. If the speed of the electrons is increased, which of the following statements is correct ?

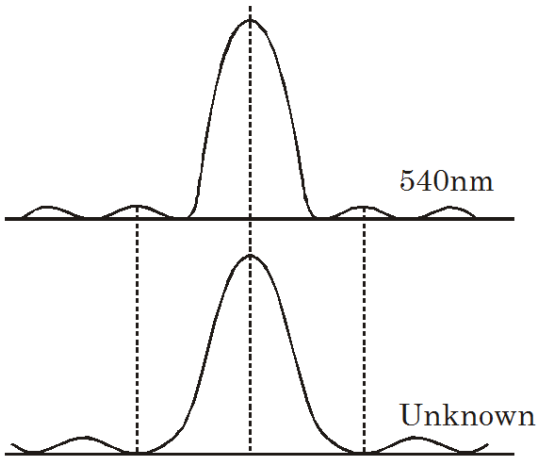
(1) The angular width of central maximum will be unaffacted.

(2) Diffraction pattern is not observed on the screen in the case of electrons.

(3) The angular width of the central maximum of the diffraction pattern will increase.

(4) The angular width of the central maximum will decrease.

35) When light of wavelength 540 nm passes through a single slit set-up of unknown width, the diffraction pattern displays its first maximum where the first minimum of light of an unknown wavelength had been observed to fall. What is the unknown wavelength ?



(1) 360 nm

(2) 270 nm

(3) 180 nm

(4) 810 nm

36) The two slits at a distance of 1 mm are illuminated by the light of wavelength 6.5×10^{-7}m. The interference fringes are observed on a screen placed at a distance of 1m. The distance between third dark fringe and fifth bright fringe will be

(1) 0.65 mm

(2) 1.63 mm

(3) 3.25 mm

(4) 4.88 mm

37) A light whose electric field vectors perpendicular to the plane of incidence are completely removed by using a good Polaroid, allowed to incident on the surface of the prism at Brewster's angle.

AIM : FREE EDUCATION TO ALL APUL

Choose the most suitable option for the phenomenon related to the prism.

(1) Reflected and refracted rays will be perpendicular to each other

(2) Wave will propagate along the surface of prism

(3) No refraction, and there will be total reflection of light**.**

(4) No reflection and there will be total transmission of light.

38) In YDSE, intensity of light passes through both slit is equal. A 100% transparent thin film is placed infront of one of the slits. Now the intensity on the centre becomes 25% of the previous intensity. The wavelength of light is 4000 Å and refractive index 1.5. The minimum thickness of the glass slab is :-

(1) 1.23×10^{-7} m

(2) 2.67×10^{-7} m

(3) 4×10^{-7} m

(4) None of these

39) In a double slit experiment $\lambda = 405$ nm, the slit separation is 20 µm and slit width is 4 µm. Number of bright fringes within central peak of diffraction envelop is :-

(1) 6

(2) 8

(3) 10

(4) 12

40) Two polaroids are crossed to each other. When one of them is rotated through 60°, then what percentage of the incident unpolarised light (falling on 1st polaroid) will be transmitted by the polaroids ?

(1) 50%

(2) 75%

(3) 62.5%

(4) 37.5%

41) In YDSE, 60 fringes are observed on the screen with light of wave length 4000 Å. The number of fringes that will be observed on the same screen with the light of wavelength 6000 Å is :-

(1) 40

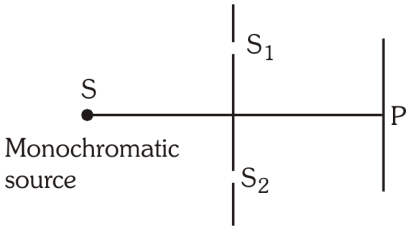
(2) 60

(3) 90

(4) cannot be obtained

AIM : FREE EDUCATION TO ALL ♥ APUL

AIM : FREE EDUCATION TO ALL ♥ APUL



42) A Young's double slit apparatus is shown in figure :-

	Column-I		Column-II
(A)	Source S move upwards	(P)	Central fringe moves upwards
(B)	A thin glass slab is placed in front of slit S_2	(Q)	Central fringe moves downward
(C)	Whole apparatus is immersed in water	(R)	Fringe width remains constant
(D)	Diameter of slit S_2 is doubled	(S)	Fringe width decreases
		(T)	Intensity of central maxima increases

(1) A→P, R ; B → Q, R ; C → R, T ; D → R, T

(2) A→Q, R ; B → Q, R ; C → S ; D → R, T

(3) A→P, R ; B → P, R ; C → S ; D → Q, R

(4) A→Q, R ; B → P, R ; C → R, T ; D → R, T

43) The main difference between the phenomena of interference and diffraction is that :

(1) Diffraction is due to interaction of light from the same wave front, whereas interference is the interaction of waves from two separate sources.

(2) Diffraction is due to superposition of light from the same wavefront, whereas interference is due to two waves derived from the same source.

(3) Diffraction is due to two waves derived from the same source, while interference is superposition of light from same wavefront.

(4) In both diffraction and interference, there are two coherent sources.

44) In a Fraunhofer diffraction at a single slit of width d with incident light of wavelength 5500 Å, the first minimum is observed, at angle of 30°. The first secondary maximum is observed at an angle θ, equal to :-

(1) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(2) $\sin^{-1}\left(\frac{1}{4}\right)$

(3) $\sin^{-1}\left(\frac{3}{4}\right)$

(4) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

45) Plane wavefront is incident on a single slit of width 0.2 mm by which Fraunhoffer diffraction pattern is formed by lens of focal length 1 m. If the second minima is formed at a distance of 3 mm from the centre then find the wavelength of light.

(1) 1500 Å
(2) 3000 Å
(3) 4500 Å
(4) 6000 Å

AIM : FREE EDUCATION TO ALL ❤ APUL

AIM : FREE EDUCATION TO ALL ♥ APUL

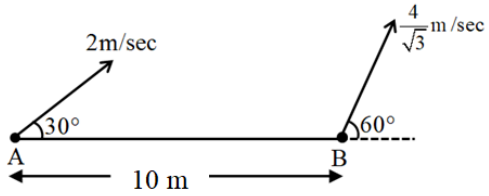
PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**1**	**3**	**2**	**4**	**4**	**1**	**2**	**4**	**1**	**2**	**1**	**4**	**2**	**1**	**3**	**3**	**3**	**1**	**1**	**1**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**3**	**2**	**4**	**2**	**1**	**3**	**2**	**2**	**1**	**1**	**4**	**4**	**3**	**4**	**4**	**2**	**4**	**2**	**3**	**4**
Q.	41	42	43	44	45															
A.	**1**	**2**	**2**	**3**	**2**															

AIM : FREE EDUCATION TO ALL ♥ APUL

PHYSICS

1)



$$\omega_{rel} = \frac{\text{relative velocity perpendicular to line joining}}{\text{separation between them}}$$

$$\omega_{rel} = \frac{\frac{4}{\sqrt{3}}\sin 60^o - 2\sin 30^o}{10} = \frac{\frac{4}{\sqrt{3}}.\frac{\sqrt{3}}{2} - 2.\frac{1}{2}}{10}$$

$$\omega_{rel} = \frac{2-1}{10} = 0.1 \text{ rad/sec}$$

2) $K = As^2$

$$\frac{1}{2}mv^2 = As^2$$

$$v^2 = \frac{2As^2}{m} \Rightarrow v = \sqrt{\frac{2A}{m}}\, s$$

$$a_t = v\frac{dv}{ds} = \sqrt{\frac{2A}{m}}\, s\sqrt{\frac{2A}{m}} = \frac{2A}{m}s$$

$$a_r = \frac{v^2}{r} = \frac{2A}{m}\frac{s^2}{r}$$

$$a_{net} = \sqrt{a_t^2 + a_r^2} = \frac{2A}{m}s\sqrt{1 + \frac{s^2}{r^2}}$$

$$F_{net} = ma_{net} = 2As\sqrt{1 + \frac{s^2}{r^2}}$$

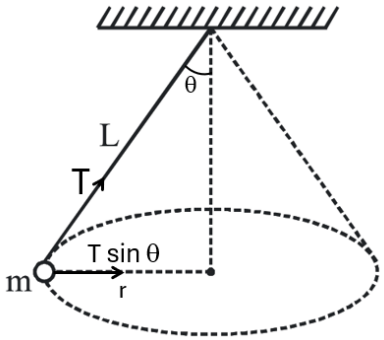
3) $u_x = \sqrt{g\ell}$

$$T = \sqrt{\frac{2 \times 2\ell}{g}} = \sqrt{\frac{4\ell}{g}}$$

$$R = u_x T = \sqrt{g\ell}\sqrt{\frac{4\ell}{g}} = \sqrt{4\ell^2} = 2\ell$$

4)

AIM : FREE EDUCATION TO ALL ♥ APUL



Here r = Lsinθ

$F_C = mr\omega^2$

$T \sin\theta = m(L \sin\theta)\omega^2$

$T = mL\omega^2$

$T_{max} = mL\omega_{max}^2$

Given m = 0.5 kg , L = 0.5 m, T_{max} =324 N.

$\omega^2 = \frac{T_{max}}{mL} = \frac{324}{0.5 \times 0.5} = \frac{324}{0.25} = 1296$

$\omega = \sqrt{1296} = 36$ rad/s

5) $\tan\theta = \frac{v_1^2}{Rg} = \frac{v_2^2}{4Rg} \Rightarrow v_2 = 2v_1$

$\left(\frac{v_2}{v_1} - 1\right) = 1$

% increase = 100%

6) As $v^2 = gr\left(\frac{\mu + \tan\theta}{1 - \mu\tan\theta}\right)$

$\Rightarrow v^2 = 9.8 \times 1000 \times \left(\frac{0.5 + 1}{1 - 0.5 \times 1}\right)$

$\Rightarrow v = 172$ m/s

7)

$\omega^2 = \omega_0^2 + 2\alpha\theta$.

After 36 rotations its angular speed becomes half

$\left(\frac{\omega}{2}\right)^2 = \omega^2 - 2\alpha\theta_1$

$\frac{\omega^2}{4^2} = \omega^2 - 2\alpha\theta_1$

$\frac{3}{4}\omega^2 = 2\alpha\theta_1$ ...(i)

For coming to rest from $\frac{\omega}{2}$:

$0 = \frac{\omega}{4^2} - 2\alpha\theta_2$

$\frac{\omega^2}{4} = 2\alpha\theta_2$ ...(ii)

From equation (i) and (ii)

$3(2\alpha\theta_2) = 2\alpha\theta_1$
$3(2\pi n_2) = 2\pi n_1$
$n_2 = \frac{n_1}{3} = \frac{36}{3} = 12$ rotations

8) $T = mg + m\omega^2 r = m\{g + 4\pi^2 n^2 r\}$
$= m\left\{g + \left(4\pi^2\left(\frac{n}{60}\right)^2 r\right)\right\} = m\left\{g + \left(\frac{\pi^2 n^2 r}{900}\right)\right\}$

9) $\frac{1}{2}m(sgR) = mgR + \frac{1}{2}mv_P^2$
$v_P = \sqrt{6gR}$

10)

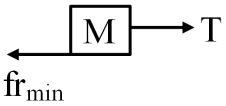
NCERT-XI, Pg. # 81

11)

$a_c = \frac{v^2}{r}$
$\Rightarrow \frac{2}{r^3} = \frac{v^2}{r}$
$\Rightarrow v = \frac{\sqrt{2}}{r}$
$p = mv$
$p = \frac{\sqrt{2}m}{r}$

12) Velocity of the block m at the lowest point of the circular path will be $v = \sqrt{2g\ell}$.

Tension at lowest point = $T = mg + \frac{mv^2}{\ell} = 3mg$
$fr_{min} = 3mg$
$\mu_{min}Mg = 3mg$
$\mu_{min} = \frac{3m}{M}$



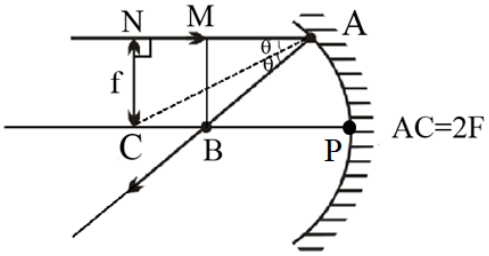
13)

A real image can only be formed by converging mirrors i.e. concave mirrors. For concave mirrors, real images form on the same side as the object. For real objects, convex mirror always produce virtual image behind the mirror.

14)

AIM : FREE EDUCATION TO ALL ♥ APUL

This question is based on marginal ray focal length



Δ CNA

$\sin\theta = \frac{f}{2f} = \frac{1}{2}$

$\theta = 30^{\circ}$

BC = AN - AM

$= \sqrt{3}f - \frac{f}{\sqrt{3}}$

$= f\left(\frac{2}{\sqrt{3}}\right)$

$\frac{BC}{f} = \frac{\frac{2}{\sqrt{3}}f}{f} = \frac{2}{\sqrt{3}}$

15)

For concave mirror

u = –30 cm, $-\frac{v}{u} = -2$, v = –60 cm

Now, $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{-60} + \frac{1}{-30} = -\frac{1}{20}$

f = –20 cm

For another location of object (at a distance 'x' cm from pole of mirror)

u = –x, $-\frac{v}{u} = 2$, v = 2x

again, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{2x} + \frac{1}{-x} = \frac{1}{-20}$

$\Rightarrow$ x = 10 cm

$\Rightarrow$ u = –10 cm

16)

f = 25 cm

$m = \frac{f}{f-u}$

$m = \frac{25}{25-(-25)} = 0.5$

17)

Wavelength of light :

AIM : FREE EDUCATION TO ALL ♥ APUL

in $X = \frac{\lambda}{1.5}$, in $Y = \frac{\lambda}{\mu}$, in $Z = \frac{\lambda}{1.6}$

(λ is wavelength in air)

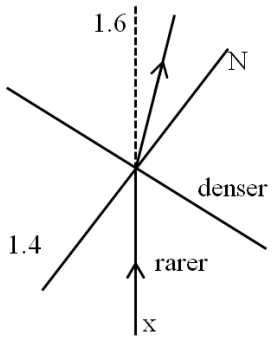
Given : Number of wavelengths in X = number of wavelengths in Y + number of wavelengths in Z

$$\Rightarrow \frac{t}{\lambda/1.5} = \frac{t/3}{\lambda/\mu} + \frac{2t/3}{\lambda/1.6}$$

$$1.5 = \frac{\mu}{3} + \frac{3.2}{3}$$

$$\Rightarrow \frac{\mu}{3} = 1.5 - \frac{3.2}{3} = \frac{1.3}{3} \Rightarrow \mu = 1.3$$

18)

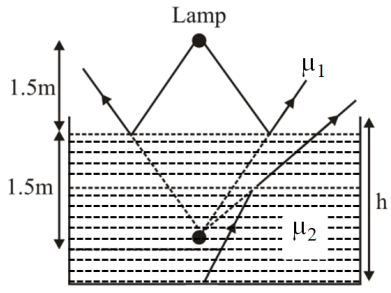


Light is going from rarer to denser medium, it should bend towards the normal, white entering denser medium.

19)

If first and last medium are same, then incident and emergent rays are parallel.

20)



$\mu = \frac{4}{3}$

d = h, d = 1.5 m,

$$\frac{d'}{d} = \frac{\mu_1}{\mu_2} \Rightarrow \frac{1.5}{h} = \frac{1}{4/3}$$

$\Rightarrow$ h = 2m

21) $V_{air} = \frac{x}{t_1}$ & $V_{med} = \frac{10x}{t_2}$

$$\sin\theta_C = \frac{\mu_{air}}{\mu_{med}}$$

$$\Rightarrow \sin\theta_C = \frac{V_{med}}{V_{air}}$$

AIM : FREE EDUCATION TO ALL ♥ APUL

$\Rightarrow \sin\theta_C = \frac{10t_1}{t_2}$

22) **Given:**
Angle of incidence (i) = 60°
Prism angle (A) = 30°
Angle between incident and emergent rays (this is the deviation,δ) = 30°.
Using the formula A + δ = i + e
30° + 30° = 60° + e
⇒ e = 0°.
For a prism, $A = r_1 + r_2$.
Since $r_2 = 0°$, $A = r_1$.
Therefore, $r_1 = 30°$.
Now, $\mu = \frac{\sin i}{\sin r_1}$
$= \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{3}$

23) $\cot(A/2) = \frac{\sin\left(\frac{A+\delta_{min}}{2}\right)}{\sin(A/2)}$
$\cos(A/2) = \sin\left(\frac{A+\delta_{min}}{2}\right)$
$\sin(90° - A/2) = \sin\left(\frac{A+\delta_{min}}{2}\right)$
$90° - A/2 = \frac{A+\delta_{min}}{2}$
$\delta_{min} = 180° - 2A$

24) After first refraction image will be formed at v_1
$\frac{\mu_2}{v_1} - \frac{\mu_1}{-\infty} = \frac{\mu_2 - \mu_1}{R}$ ....(i)
Final image (after second refraction) will be formed at v_2
$\frac{\mu_3}{v_2} - \frac{\mu_2}{v_1} = \frac{\mu_3 - \mu_2}{R}$ ....(ii)
Adding (i) & (ii)
$\frac{\mu_3}{v_2} = \frac{\mu_3 - \mu_1}{R} \Rightarrow v_2 = \frac{\mu_3 R}{\mu_3 - \mu_1}$
 focal length will be = $\frac{\mu_3 R}{\mu_3 - \mu_1} = \frac{2R}{(2-1)} = 2R$
 n = 2

25)

If we place concave lens of focal length 60 cm in contact with previous lens, then :-
$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} \Rightarrow \frac{1}{f_{eq}} = \frac{1}{20} + \frac{1}{(-60)} = \frac{1}{30}$

AIM : FREE EDUCATION TO ALL APUL

f_{eq} = 30 cm
Now, point object will be at focus of lens combination whose image will form at ∞.

26)

Power: $P = \left(\frac{n_{lens}}{n_{medium}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$
In air $n_{lens} > 1$ (converging). In water, lens becomes diverging, so $n_{lens} < 1.33$. Also, $n_{lens} > 1$
Thus: n_{lens} will be between unity and 1.33.

27)

The total power of the thin lens is

$$P = P_1 + P_2 = \frac{\mu_2 - \mu_1}{R_1} + \frac{\mu_3 - \mu_2}{R_2}$$

$$P = \left(\frac{1.2 - 1.3}{10} + \frac{1.5 - 1.2}{-10}\right) \times 100$$

= (–0.1 – 0.3) × 10
= – 4D

28)

In the displacement method, for magnification m and displacement d between the two lens positions, the focal length is

$$f = \frac{md}{m^2 - 1}$$

Given
m = 3, d = 24 cm

$$\Rightarrow f = \frac{md}{m^2 - 1} = \frac{3 \times 24}{3^2 - 1} = \frac{72}{9 - 1} = 9 \text{ cm}$$

29)

f_0 = 2 cm , f_e = 5 cm, u_0 = –3 cm

$$\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$$

$$\frac{1}{2} = \frac{1}{v_0} - \frac{1}{(-3)}$$

v_0 = 6cm

$$\frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e}$$

$$\frac{1}{5} = \frac{1}{(-25)} - \frac{1}{u_e}$$

$$u_e = -\frac{25}{6} = -4.16\text{cm}$$

length of tube = $v_0 + |u_e|$ = 6 + 4.16 = 10.16 cm

AIM : FREE EDUCATION TO ALL ♥ APUL

AIM : FREE EDUCATION TO ALL ♥ APUL

30)

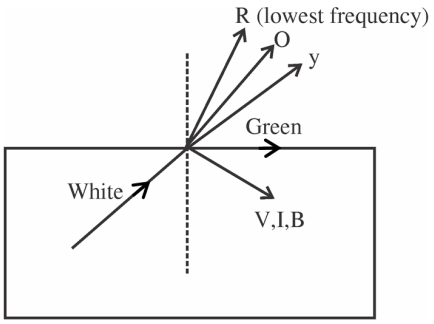
Given,
Magnifying Power (MP) = 5
Separation (d) = 36 cm

$$MP = 5 = \left|-\frac{f_0}{f_e}\right| \Rightarrow f_0 = 5f_e$$

$L = 36 = f_0 + f_e$
$36 = 5\, f_e + f_e$
$6f_e = 36$
$f_e = 6$ cm
$f_0 = 5f_e$
$= 5 \times 6 = 30$ cm

31) $m = \frac{f}{f-u}$
When m = +3,
$3 = \frac{-15}{-15-u} \left(\because f = \frac{R}{2}\right)$
–15 – u = –5
– u = –5 + 15
$\boxed{u = -10\,cm}$
When m = –3,
$-3 = \frac{-15}{-15-u} \Rightarrow -15-u = 5$
–u = 5 + 15
u = –20 cm

32)



33)

$m = \frac{f}{f+u}$

$m_{25} = \frac{20}{20-25}$

$m_{50} = \frac{20}{20-25}$

$\frac{m_{25}}{m_{50}} = 6$

34) As speed of electrons is increased so wavelength of electrons will decreases. Therefore the angular width ($\propto \lambda$) of the central maximum of diffrection pattern will decrease.

35) $\frac{3D\lambda_1}{2a} = \frac{D\lambda_2}{a}$

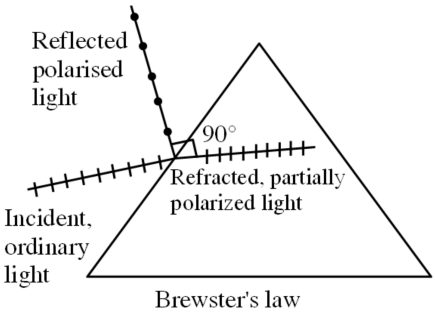
$\lambda_2 = \frac{3 \times 540}{2}$ = 810 nm

36) $\Delta y = y_{5B} - y_{3D}$

$= \frac{5\lambda D}{d} - \frac{5}{2}\lambda\frac{D}{d} = \frac{5}{2}\lambda\frac{D}{d}$

= 1.625 mm

37)



Brewster's law

But as in the incident light perpendicular electric field vectors are completely removed so there will be no reflection and there will be total transmission of light, explained by an experiment in NCERT.

[Reference old NCERT Part-2 Pg-380, (A special case of total transmission)]

Note : Since direction of polarization is not mentioned hence most suitable option (4) corresponding to case in which electric field is absent perpendicular to plane consisting incident and normal.

38) $I_1 = I_2 = I'$

Maximum Intensity = 4I'

$I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$

$I' = I' + I' + 2\sqrt{I'}\sqrt{I'}\cos\phi$

$I' = 2I' + 2I'\cos\phi$

$1 = 2(1 + \cos\phi)$

$\frac{1}{2} = 1 + \cos\phi$

$\cos\phi = -\frac{1}{2}$

$\phi = 120°$ or $\frac{2\pi}{3}$

$\Delta x = \frac{\lambda}{3}$

$\frac{\lambda}{3} = (\mu - 1)t$

AIM : FREE EDUCATION TO ALL ♥ APUL

$\frac{4000 \times 10^{-10}}{3} = (1.5 - 1)t$

$t = 2.67 \times 10^{-7}$ m

39) $\lambda = 405 \times 10^{-9}$ m, $a = 4 \times 10^{-6}$ m,

$d = 20 \times 10^{-6}$ m

$\Rightarrow \beta_d = n\beta i \quad ; \quad \frac{2\lambda D}{a} = n\frac{\lambda D}{d}$

$n = \frac{2d}{a} = 10$

40) The intensity of light emerging from the first polaroid is $I_1 = \frac{1}{2}I_0$

The intensity of light emerging from the second polaroid is $I_2 = I_1 \cos^2\theta$

$I_2 = I_1 \cos^2 30^\circ \Rightarrow \frac{I_2}{I_0} = \frac{3}{8}$

 transmission percentage

$= \frac{I_2}{I_0} \times 100 = \frac{3}{8} \times 100 = 37.5\%$

41) $n_1\lambda_1 = n_2\lambda_2$

$60 \times 4000Å = n_2 \times 6000Å$

$\frac{60 \times 4}{6} = n_2$

$40 = n_2$

42) (A) Due to change in path difference so shift is in opposite direction to the source movement

(B) Due to glass slab optical path length increases

(C) Due to change in refractive index λ changes

(D) Due to increase in diameter intensity increases

43)

Remember: Diffraction involves waves from the same single wavefront (like a single slit), while interference involves two waves from separate coherent sources (like double slits).

44) $a \sin\theta = \lambda$

$a \sin\theta' = \frac{3\lambda}{2a}$

$\sin\theta' = 3/2 \times 1/2 = 3/4$

$\theta' = \sin^{-1}(3/4)$

45) $x = \frac{n\lambda D}{d} \Rightarrow 3 \times 10^{-3} = \frac{2 \times \lambda \times 1}{0.2 \times 10^{-3}}$

AIM : FREE EDUCATION TO ALL ♥ APUL

$\lambda = 3 \times 10^{-7} m = 3000 \text{ Å}$

AIM : FREE EDUCATION TO ALL ❤ APUL